Code No: 155AM **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
COMPUTER GRAPHICS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Write Bresenham's line drawing algorithm and explain with suitable example.
b) Describe about different video-display devices. [7+8]

2.a) Explain about Bresenham's ellipse drawing algorithm.
b) Describe about scan line polygon filling algorithm. [8+7]

3.a) Summarize about Translation and scaling in 2-D geometrical transforms.
b) Write about Cohen-Sutherland line clipping algorithm with an example. [7+8]

4.a) Describe about viewing coordinate reference frame with an example.
b) Determine the rotation of a straight line (x1,y1) to (x2,y2) with respect to the point (1, 2). [8+7]

5.a) Compare polygon surfaces with quadric surfaces.
b) Describe about different polygon rendering methods. [7+8]

6.a) Describe about 3-D objects representation using B-spline curves.
b) Compare the different illumination models. [8+7]

7.a) Describe about 3-D rotations with suitable examples.
b) Describe about composite transformations and its matrices. [7+8]

8.a) Explain about design of animation sequence with an example.
b) Describe about different motion specification methods. [7+8]

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USED PAPERS SEP-2021